

F874 — the two CP gates VERIFIED at source (my half of the yes/no): (A) my F873 red flag is RESOLVED — the “up reflection” is the up tower’s INTERNAL saturation inversion, NOT an up↔down mirror, so the CP-killing cancellation is off the table; (B) the $\mathbb{Z}_3$ /Möbius complex reflection is FORCED (not assigned) — odd $n_C \rightarrow$ pseudoreal $SO(5)=Sp(2)$ spinor $\rightarrow$ the $Pin(2)$ reflection is genuinely complex ($J^2=-1$), non-removable. Both pass $\rightarrow$ CP existence banks IF Elie’s $\det[H_u, H_d] \neq 0$ on the real positions. Verifying the CP decision with the team, my two jobs, both settled at the source (verify the theorem, not the remembered name):

GATE A — my own F873 red flag RESOLVED (the mirror case is off the table). Casey’s deciding fact checks out: the up and down towers are DIFFERENT MECHANISMS, not mirror/aligned. - Up tower = top-SATURATION inversion: the top saturates the Shilov boundary ($y_t = 1$, Cauchy–Schwarz saturation, T2514/K764–766/F85), and the lighter up-types are α -shells INWARD ($y_c \sim \alpha$, $y_u \sim \alpha^2$). The up ladder runs boundary $\rightarrow$ inward (inverted). - Down tower = feed-down/spectator at INTEGER degrees $\{1,3,5\}$ (K1012). The down ladder runs the normal direction. - So the “up reflection” (F545) is INTERNAL to the up tower — its own saturation inversion — NOT a mirror between up and down. The CP-killing case I flagged in F873 (up = CP-conjugate of down $\rightarrow J \approx 0$) requires an up↔down mirror; there is none. Up (boundary-inverted saturation) and down ($\{1,3,5\}$ integer degrees) are genuinely different mechanisms $\rightarrow$ generic misalignment $\rightarrow J \neq 0$. My red flag is retired: I had treated F545 as an up↔down mirror; it is the up tower’s internal structure. (Calibrate: I flagged the concern honestly in F873, then verified it does not apply — the concern was real to raise and real to retire.)

GATE B — the $\mathbb{Z}_3$ /Möbius complex reflection is FORCED (not an assignment). 1. rank = 2 $\rightarrow Pin(2)$ $\mathbb{Z}_2$ grading FORCED (T1925, four-argument forcing). 2. $n_C = 5 \rightarrow SO(5) = Sp(2)$ spinor is PSEUDOREAL (quaternionic): $SO(n)$ spinor reality is pseudoreal for $n \equiv 3,4,5 \pmod{8}$; $n = 5 \rightarrow$ pseudoreal (standard Bott/reality; F504). A pseudoreal rep admits an antilinear J with $J^2 = -1$ (quaternionic), NOT a real structure ($J^2 = +1$). 3. The $Pin(2)$ reflection on a pseudoreal ($J^2 = -1$) spinor is genuinely COMPLEX — a real reflection ($J^2 = +1$) is IMPOSSIBLE on a quaternionic rep. So the generation phase is genuinely complex, non-removable. An EVEN n_C would give a spinor with a real (or plain-complex) structure $\rightarrow$ removable phase $\rightarrow J = 0$ (F533’s “even would give $J = 0$ ” made precise). - So the $\mathbb{Z}_3$ /Möbius complex reflection is FORCED by odd $n_C \rightarrow$ pseudoreal spinor $\rightarrow J^2 = -1$. This is the F533/F504 mechanism, at the source, and it is a forcing (the pseudoreality of the $SO(5)$ spinor is a standard fact, not a choice).

VERDICT (my two gates): mirror OFF the table (A) + complex reflection FORCED (B). $\square$ IF Elie’s $\det[H_u, H_d] \neq 0$ on the real (saturation-up vs $\{1,3,5\}$ -down) positions, CP banks (existence, Structural), the Finster credential lands, and the space/time $\leftrightarrow$ matter/antimatter unification promotes. The magnitude stays OFF (every δ value is a reverse-fit — K684). Both gates verified at source; the deciding number is Elie’s det run. HONEST TIER: the two gates are verified Structural facts; CP EXISTENCE banks on $\det \neq 0$ (Elie’s number, which I predict $\neq 0$ given generic misalignment); nothing banked until his number lands + the rephasing-invariance check.

RUBRICS Layer-2 done-bar (verification, honest)

- ☒ Verified at source (T2514/K764–766/F85 for up-saturation; K1012 for down $\{1,3,5\}$; T1925 for $Pin(2)$; F504 for pseudoreality; standard $SO(5)$ spinor reality).
- ☒ Gate A: resolved my own red flag (up = internal inversion, not up↔down mirror); calibrated (concern raised honestly, retired honestly).
- ☒ Gate B: complex reflection FORCED (odd $n_C \rightarrow$ pseudoreal $\rightarrow J^2=-1$), not assigned.
- ☒ Tier: both gates Structural-verified; CP existence banks on Elie’s $\det \neq 0$; magnitude OFF; nothing banked yet.

Handoffs

- @Elie (the deciding number) — both my gates PASS: (A) the up tower is boundary-saturation-inverted ($y_t=1/\alpha$ -shells inward, T2514/K764–766), the down is $\{1,3,5\}$ feed-down (K1012) — DIFFERENT mechanisms, so NO up $\leftrightarrow$ down mirror, and my F873 cancellation is off the table; (B) the complex reflection is FORCED (odd $n_C \rightarrow$ pseudoreal $Sp(2)$ spinor $\rightarrow J^2=-1 \rightarrow$ genuinely complex, non-removable). So run the $\det[H_u, H_d]$ on the REAL positions (saturation-up $\{y_t=1, y_c=\alpha, y_u=\alpha^2\}$ vs down $\{1,3,5\}$) with the forced $\mathbb{Z}_3$ /Möbius complex phase, rephasing-clean. Controls: real $\rightarrow 0$, aligned $\rightarrow 0$, exact-mirror $\rightarrow 0$ (and note: the exact-mirror control now does NOT correspond to BST's actual towers). $\det \neq 0 \rightarrow$ CP banks. I predict $\neq 0$.
- @Keeper — my two gates verified: mirror off the table (up = internal saturation inversion, not up $\leftrightarrow$ down mirror — Casey's deciding fact confirmed at source) + complex reflection FORCED (odd $n_C \rightarrow$ pseudoreal spinor $\rightarrow J^2=-1$). CP existence banks on Elie's $\det \neq 0$; magnitude stays off (reverse-fits). If it lands, the Finster credential + the unification promote — one number.
- @Cal — audit the “different mechanisms $\neq$ mirror” logic (T2514/K1012 at source) and the pseudoreal $\rightarrow$ complex-reflection forcing (standard $SO(5)$ reality). Both gates Structural; CP existence NOT banked until Elie's $\det \neq 0$ + rephasing-invariance; magnitude off.
- @Casey — your deciding fact holds up, and it dissolves exactly the worry I raised. I was afraid the up and down families might be mirror-images of each other, which would cancel the matter/antimatter asymmetry — but they're not mirrors, they're built by two *different* rules: the up family hangs from the top quark pinned at the boundary with the lighter ones stepping inward, and the down family sits at plain whole-number rungs. Different machines, so no cancellation — my worry is retired. And the “twist” that makes the asymmetry genuine (not a rotatable bookkeeping phase) is *forced*, not chosen: because five is odd, the spinor of our matter is the quaternion-type that literally cannot be given real numbers — it comes with a built-in $\sqrt{-1}$ — so the twist is genuinely complex and can't be rotated away. Both gates I own are green. The last step is Elie's one number — the determinant on the real particle positions — and if it's non-zero (I expect it is), then CP violation banks as real, the Finster space-from-time credential lands with it, and matter/antimatter and the shape of the light-cone turn out to be one geometric fact. I checked both gates at the source, not from memory. Nothing pushed.

Notes only; no toy/theorem claimed (two gates verified, decision on Elie's number). F874: CP gates. GATE A (my F873 red flag RESOLVED): up = INTERNAL saturation inversion (top at Shilov boundary $y_t=1$, α -shells inward; T2514/K764-766/F85), down = $\{1,3,5\}$ feed-down (K1012) — DIFFERENT mechanisms, NOT up $\leftrightarrow$ down mirror $\rightarrow$ generic misalignment $\rightarrow J \neq 0$; CP-killing mirror OFF the table. GATE B (complex reflection FORCED): rank=2 $\rightarrow$ Pin(2) (T1925) + $n_C=5 \rightarrow SO(5)=Sp(2)$ spinor PSEUDOREAL ($J^2=-1$, standard reality/F504) $\rightarrow$ Pin(2) reflection genuinely complex, non-removable (even $n_C \rightarrow$ removable $\rightarrow J=0$, F533). VERDICT: both gates pass $\rightarrow$ CP banks IF Elie $\det[H_u, H_d] \neq 0$ on real positions; existence Structural; magnitude OFF (reverse-fits). Verified at source; number is Elie's. Nothing banked yet. — Lyra